

# One-For-All, TruthX and SLiMTruth

## A Study on Transferability between LLMs for Hallucination Detection and Mitigation

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# LLMs and Hallucinations

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- **LLM**: probabilistic models that predict the next token
- **Hallucination**: generation of syntactically correct but **factually incorrect** text
- Limits usage in critical domains (medicine, law)
- **Goal**: *Detect* and *mitigate* hallucinations in a transferable way across different models

## Example

**User:** How many r's in strawberries?

**LLM:** 2

# Platonic Representation Hypothesis

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## Hypothesis (Huh et al., 2024)

Different AI models, trained on different data and objectives, converge towards a **shared statistical representation of reality**.

- LLMs from different families (Llama, Gemma) may share a **common internal representation** of truthfulness
- If true, it is possible to:
  - Transfer **probers** between models without retraining
  - Transfer **steering vectors** for mitigation

*This thesis explores transferability for hallucination detection and mitigation*

# Experimental Setup

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## Models:

- Llama-3.1-8B-Instruct
- Gemma-2-9B-IT

## Datasets:

- **BeliefBankFacts**: simple facts
- **BeliefBankConstraints**: logical constraints
- **HaluEval**: complex contexts

## Two tasks:

1. **Detection**: detect *when* the model hallucinates
2. **Mitigation**: *reduce* hallucinations during generation

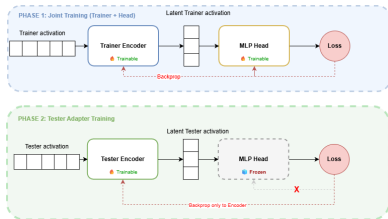
# Detection: One-For-All

## Idea:

- Each model has a model-specific **Encoder**
- A single shared **Classification Head** (frozen)
- The encoder learns to project activations into a common space

## Training:

1. Train Encoder + Head on the **Trainer**
2. Freeze the Head
3. Train only the **Tester's** Encoder



# Mitigation: TruthX

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**Idea** (Zhang et al., 2024):

- Autoencoder learns two spaces:
  - **Semantic space**: content
  - **Truthfulness space**: veracity
- During inference: modify activations towards the “truthful” space

**Cross-model transfer:**

- Procrustes alignment between latent spaces

## Advantages

- Contrastive decomposition
- Preserves semantic content

## Disadvantages

- Can degrade if baseline already accurate

# Mitigation: SLiMTruth

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**Idea** (adaptation of SLiM):

- Modulates activations of a **single layer** via affine transformation:

$$h' = \gamma \cdot h + \beta$$

- $\gamma$  and  $\beta$  are learned for each layer
- More localized intervention compared to TruthX

**Cross-model transfer:**

- Procrustes alignment + layer mapping

## Advantages

- Very high in-domain reductions
- Up to **-89.9%** hallucination rate

## Disadvantages

- Highly layer-specific
- Poor cross-model transferability

## Detection Results: One-For-All

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| Dataset               | Trainer | Tester | Type | AUROC Tr | AUROC Te     |
|-----------------------|---------|--------|------|----------|--------------|
| BeliefBankFacts       | Gemma   | Llama  | attn | 0.992    | <b>0.997</b> |
| BeliefBankFacts       | Llama   | Gemma  | attn | 0.998    | <b>0.989</b> |
| BeliefBankConstraints | Gemma   | Llama  | attn | 0.999    | <b>0.999</b> |
| BeliefBankConstraints | Llama   | Gemma  | attn | 0.999    | <b>0.998</b> |
| HaluEval              | Gemma   | Llama  | attn | 0.768    | <b>0.855</b> |
| HaluEval              | Llama   | Gemma  | attn | 0.850    | <b>0.768</b> |

- **BeliefBank**: near-perfect transferability (AUROC > 0.99)
- **HaluEval**: complex task, but transfer still works



## Detection Results: Cross-Domain

| Train | Test | Acc Tr | Acc Te       |
|-------|------|--------|--------------|
| BBC   | HE   | 0.991  | <b>0.995</b> |
| BBF   | HE   | 0.985  | <b>0.995</b> |
| HE    | BBC  | 0.872  | 0.907        |
| HE    | BBF  | 0.877  | 0.906        |

### Key Observation

Encoders trained on **simple** tasks (BeliefBank) generalize well to **complex** tasks (HaluEval). The reverse works less well.

## Mitigation Results: Standard Setting

| Model | Dataset | TruthX  |       |               | SLiMTruth |       |               |
|-------|---------|---------|-------|---------------|-----------|-------|---------------|
|       |         | HR Base | HR    | $\Delta$      | HR Base   | HR    | $\Delta$      |
| Gemma | BBC     | 44.7%   | 41.7% | -6.6%         | 44.7%     | 18.3% | <b>-59.1%</b> |
| Gemma | HE      | 24.2%   | 15.8% | <b>-34.6%</b> | 24.2%     | 17.5% | -27.6%        |
| Llama | BBC     | 52.8%   | 49.3% | -6.8%         | 52.8%     | 25.8% | <b>-51.2%</b> |
| Llama | BBF     | 6.7%    | 28.1% | <b>+318%</b>  | 6.7%      | 0.7%  | <b>-89.9%</b> |
| Llama | HE      | 27.6%   | 13.8% | <b>-50.3%</b> | 27.6%     | 11.6% | -57.9%        |

- **SLiMTruth**: best absolute reductions (up to **-89.9%**)
- **TruthX**: can degrade on already accurate baselines (BBF)

# Mitigation Results: Cross-Model Transfer

| Train | Eval  | Dataset | TruthX |               | SLiMTruth |                |
|-------|-------|---------|--------|---------------|-----------|----------------|
|       |       |         | HR     | $\Delta$      | HR        | $\Delta$       |
| Gemma | Llama | HE      | 17.1%  | <b>-38.0%</b> | 96.3%     | <b>+248.6%</b> |
| Llama | Gemma | HE      | 15.3%  | <b>-36.7%</b> | 56.3%     | <b>+133.0%</b> |
| Gemma | Llama | BBC     | 51.3%  | -3.0%         | 49.8%     | -5.8%          |
| Llama | Gemma | BBC     | 41.3%  | -7.6%         | 70.6%     | <b>+58.1%</b>  |

## Conclusion

**TruthX** transfers better cross-model ( $\sim -37\%$  on HaluEval).  
**SLiMTruth** fails catastrophically in cross-model transfer.

# Conclusions

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## Detection (One-For-All):

- Very high transferability on factual tasks (AUROC  $>$  0.99)
- Works on complex tasks too
- Cross-domain asymmetry: simple  $\rightarrow$  complex works better

## Mitigation:

- **SLiMTruth**: best in-domain (up to -89.9%)
- **TruthX**: best cross-model ( $\sim$ -37% on HaluEval)
- Trade-off: in-domain effectiveness vs transferability

# Conclusions

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1. **One-For-All:** universal prober for detection transferable across different architectures
2. **Platonic Hypothesis Validation:** truthfulness representations are partially shared across LLMs
3. **Comparative Analysis TruthX vs SLiMTruth:** characterization of trade-offs for mitigation
4. **Limitation Identification:** vulnerability on accurate baselines, cross-model difficulties for SLiMTruth

# Future Work

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- **Adaptive steering:** modulate intensity based on model confidence
- **Improve cross-model SLiMTruth:** learn correspondences between layers
- **Multimodality:** extend to Vision-Language models

**Thank you for your  
attention!**